

Paper 18.25 - VOA / Moonshine Route Toolkit

CQE-paper-18.25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-18.25.md

Paper 18.25 - VOA / Moonshine Route Toolkit

Purpose

This supplement exposes the Paper 18 route checks. It is a reviewer tool, not the proof itself.

Digital Route

Run:

```
python production/formal-papers/CQE-paper-18/verify_voa_moonshine_routes.py
```

The receipt must include:

```
finite VOA seed
Z4 route template
Monster scalar
bounded McKay matrix bootstrap
lookup-harness deferred status
correction_via_voa open status
Monster-D4 lift bounded/open-gap status
```

Analog Route

Lay eight state cards. Mark two weight-0 cards and six weight-5 cards. Then add a four-frame route strip:

```
C frame
R frame
C-flipped frame
L frame
```

The two fixed cards remain white. The six period-4 cards move through the route strip. The Monster/McKay card is marked as bounded table evidence unless the full correction route has been supplied.

Hidden-Guess Diagnostic

Before revealing the receipt, classify each route as:

```
closed finite seed
closed static route template
bounded table evidence
bounded open-gap route
open global route
invalid promotion
```

Reveal the receipt only after the choice is made.

Boundary

This toolkit may show the finite VOA seed and bounded McKay bootstrap. It may not show `correction\_via\_voa` as complete, and it may not turn bounded tables into a full Moonshine proof.

Conclusion

Paper 18.25 makes the route visible without moving the proof boundary.